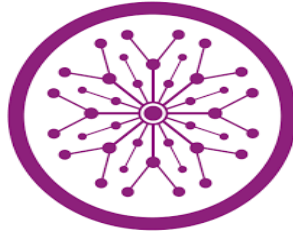


CONTEXT SWITCHING



Semester Project Report by

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Abstract

Modern operating systems rely heavily on multitasking to efficiently manage multiple processes running concurrently. A crucial component that makes this possible is context switching, the mechanism through which the CPU shifts execution from one process to another. Although this operation occurs frequently within the system, the underlying steps such as saving the current process state, updating the Process Control Block (PCB), and restoring the state of the next scheduled process often remain hidden from learners and difficult to grasp conceptually. This project aims to bridge that learning gap by developing an interactive Context Switching Simulator based on the Round Robin Scheduling (RRS) technique, one of the most widely used preemptive scheduling algorithms. The simulator visually demonstrates how processes enter the ready queue, receive CPU time through fixed time slices, and undergo state transitions during each context switch. It also displays real-time updates of PCB components such as program counter, registers, burst time, and remaining time, enabling students to observe how the CPU handles time-sharing at the hardware–software interface. Through dynamic animations, color-coded process states, and step-by-step execution, the tool converts an abstract OS concept into a clear and intuitive learning experience. Overall, the simulator enhances conceptual understanding, supports hands-on experimentation, and provides an educational platform for exploring the internal workflow of process scheduling and multitasking in operating systems.

Introduction

Operating systems must manage multiple processes efficiently to maintain system responsiveness and fairness among users. One of the most critical techniques that ensures smooth multitasking is **context switching**, in which the CPU stops the execution of one process and loads another. Every time a context switch occurs, the system must save important information about the current process such as register values, program counter, and process state into its PCB. Once this information is saved, the CPU can safely load the next process and resume execution. Despite its importance, context switching is a concept that many students struggle to grasp because it involves internal system-level operations that cannot be observed directly. Text-based explanations often fail to convey the actual flow of states, scheduling decisions, and CPU operations. For this reason, a simulation provides a powerful way to visualize how context switching works. This project focuses on creating a simulation using **Round Robin Scheduling**, a widely used preemptive scheduling algorithm that provides equal CPU time to all processes. Through this simulation, students can observe how processes enter and leave the Ready Queue, how the CPU handles time quantum expiration, and how PCBs are updated after each switch.

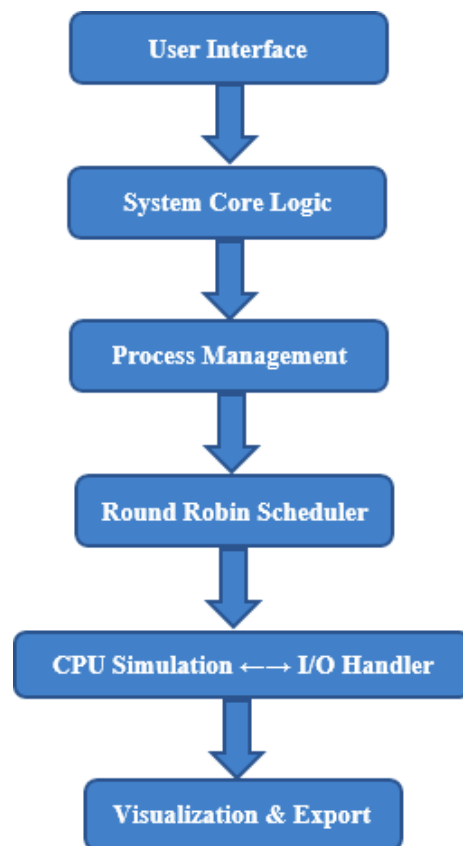


Figure 1: Block Diagram of Context Switching

Problem Statement

Understanding how context switching works in an operating system is challenging because most of the process management occurs internally and is not visible to learners. Concepts such as:

- Saving the process state
- Loading the next scheduled process
- PCB updates
- Managing Ready, Running, and Blocked states
- Round Robin Scheduling decisions

are difficult to visualize through theory alone. Students often struggle to understand how processes move between queues, how the operating system ensures fairness, and how multitasking happens at the CPU level. Therefore, there is a strong need for a **visual simulation tool** that clearly demonstrates how context switching and scheduling decisions are executed in real time.

Project aims and goals

This project is designed with the following objectives:

- To simulate the concept of context switching using the **Round Robin Scheduling** algorithm.
- To demonstrate, step-by-step, how a CPU **saves the current process state** and **loads the next process**.
- To visually represent transitions between **Ready, Running, and Blocked** states.
- To update PCB details such as program counter, CPU burst time, and process state after each switch.
- To use appropriate **queue structures** to represent the movement and management of processes.
- To provide a **clear educational tool** that helps students understand multitasking and scheduling in operating systems.

Project Methodology

The development of the Context Switching Simulator followed a structured methodology to accurately represent how an operating system handles process scheduling and context switching.

1. Requirement Analysis

The team first studied key OS concepts such as context switching, Round Robin Scheduling, process states, and the Process Control Block (PCB). The primary requirement identified was to create a visual tool that shows how processes move between Ready, Running, and Blocked states while their PCB values update in real time.

2. System Design

The simulator was designed with the following main components:

- **Process Structure & PCB:** Each process was assigned a PID, burst time, and PCB fields including program counter, registers, and state.
- **Queue System:** A Ready Queue (FIFO) was used to manage all waiting processes.
- **Scheduling Algorithm:** Round Robin Scheduling with a fixed time quantum was chosen to simulate preemptive multitasking.
- **Visual Interface:** Designed to clearly show queues, process states, and PCB updates during each switch.

3. Algorithm Development

A step-by-step algorithm was created:

1. Select the first process from the Ready Queue.
2. Load its PCB and run it for the fixed time quantum.
3. When the quantum expires, save the PCB and update burst time.
4. If the process is not finished, move it back to the end of the Ready Queue.
5. Select the next process and repeat the cycle until all processes complete.

4. Implementation

The algorithm and design were implemented into a functional simulation. The system handled process creation, time quantum execution, PCB saving/loading, queue operations, and state transitions. The visual interface updated in real time to show each switch and process movement.

5. Testing & Validation

The simulator was tested with multiple processes of different burst times to verify:

- Correct Round Robin scheduling
- Accurate PCB updates
- Proper state transitions
- Clear and synchronized visual outputs

All tests confirmed that the simulator closely resembles real OS behavior during context switching.

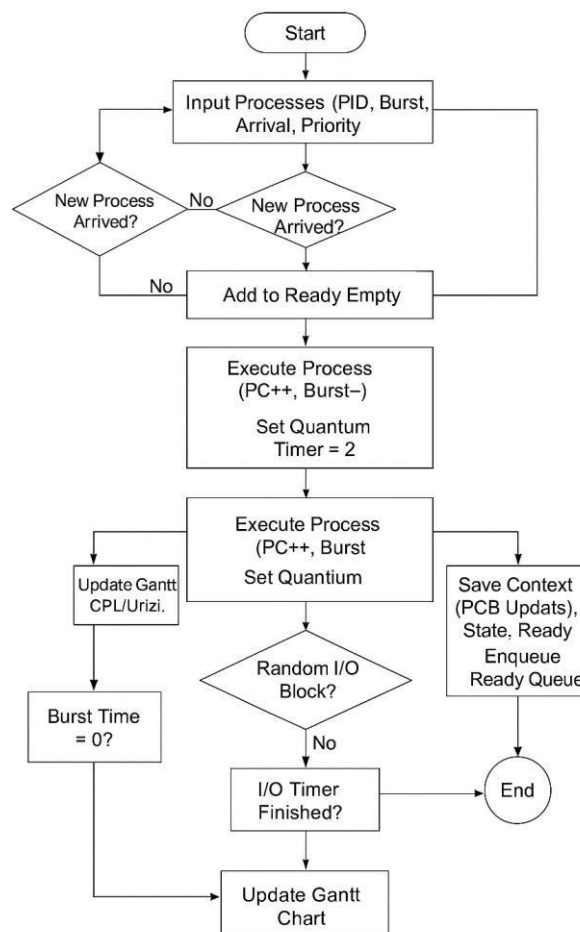


Figure 2: Flowchart Diagram of Context Switching

Conclusion

The Context Switching Simulator developed in this project successfully demonstrates how an operating system manages multitasking through the Round Robin Scheduling algorithm. By visually illustrating the transitions of processes between Ready, Running, and Blocked states and showing real-time updates of the Process Control Block (PCB), the simulator transforms a complex and hidden OS mechanism into an understandable and interactive learning experience. Through accurate queue handling, PCB management, and time-quantum-based execution, the project provides a realistic representation of how CPUs share time among multiple processes. This simulation not only deepens conceptual understanding but also bridges the gap between theoretical operating system concepts and their practical execution. The tool makes it easier for students to visualize how context switching works internally and appreciate the importance of efficient scheduling in modern operating systems. Overall, the project meets its objectives by offering an effective educational resource that enhances learning, improves clarity, and provides a strong foundation for further study in OS process management and scheduling.

Final picture of Project

Context Switching Simulator

Upload Process (Image/Video)

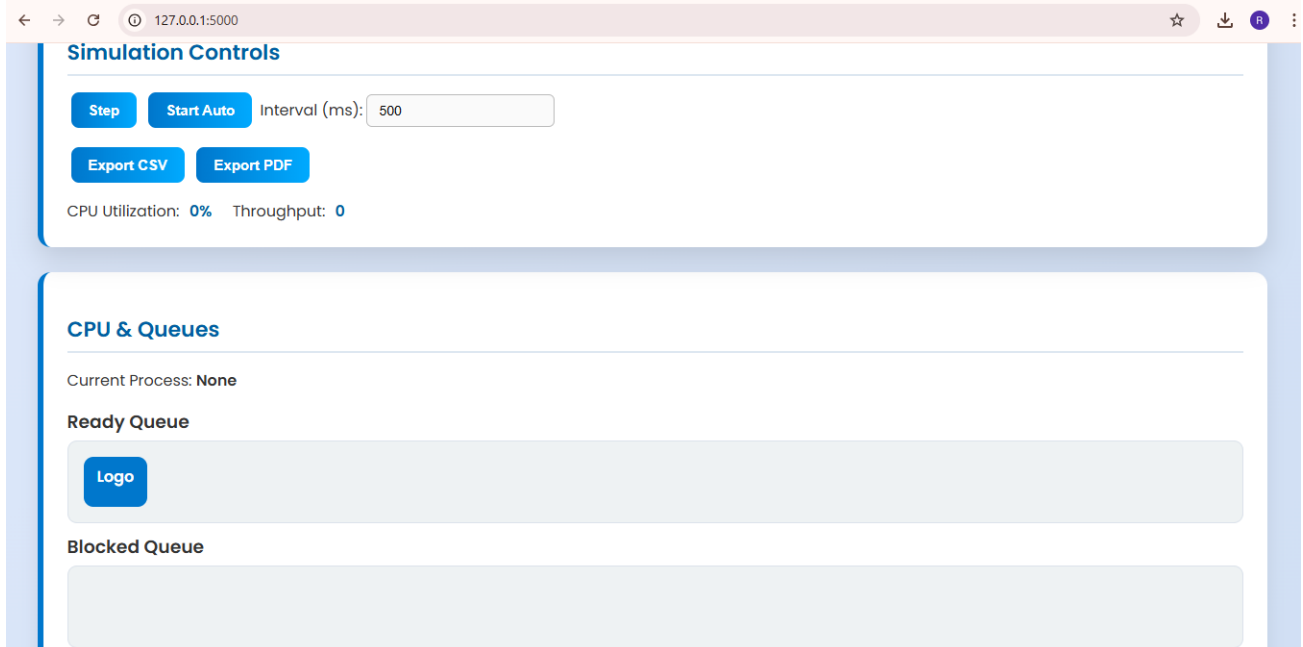
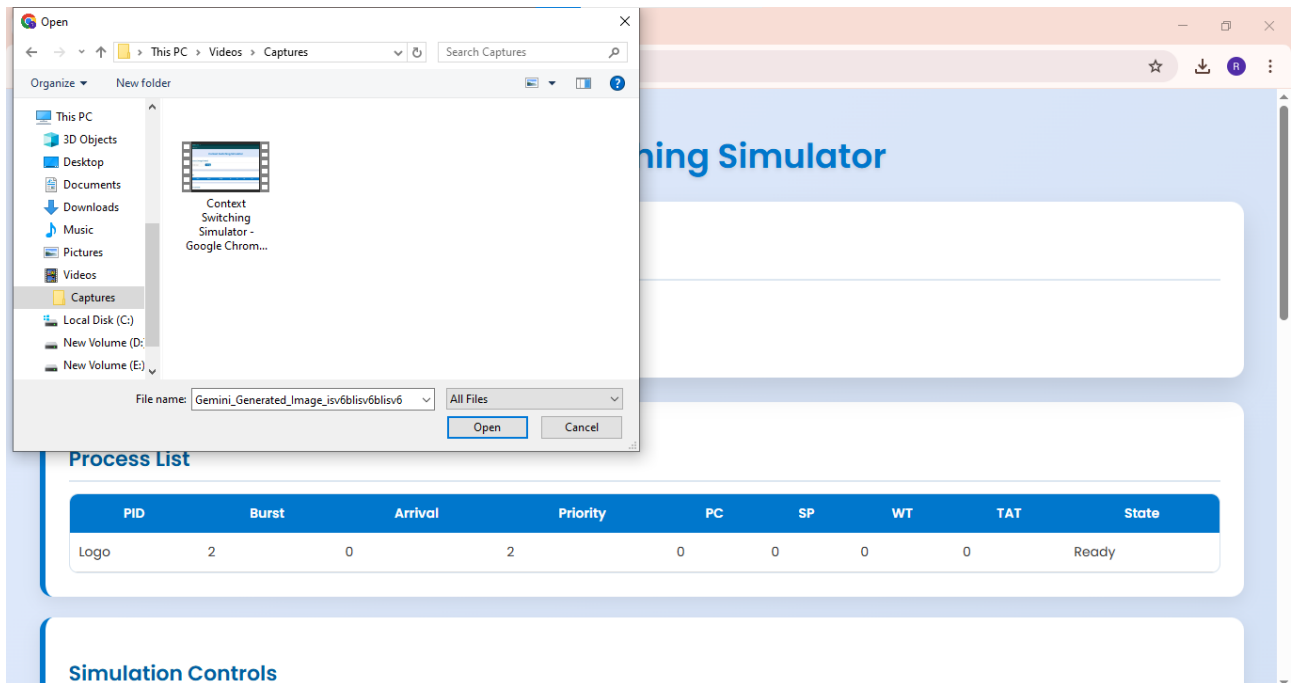
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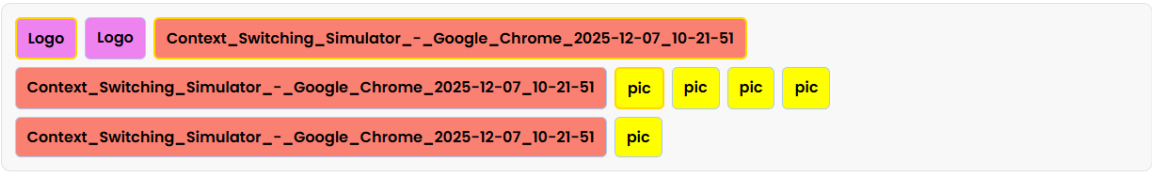
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Process List

PID	Burst	Arrival	Priority	PC	SP	WT	TAT	State
Logo	2	0	2	0	0	0	0	Ready
Context_Switching_Simulator_-_Google_Chrome_2025-12-07_10-21-51	2	1	2	0	0	0	0	New
pic	5	2	4	0	0	0	0	New



Gantt Chart



Execution History

Time	Process	State	PC	SP	Ready Queue	Blocked Queue	CPU Utilization	Throughput
0	Logo	Running	1	1			0%	0
1	Idle	Idle	0	0	Context_Switching_Simulator_-_Google_Chrome_2025-12-07_10-21-51		200%	1
2	Context_Switching_Simulator_-_Google_Chrome_2025-12-	Running	1	1	pic		150%	0.5